Problem Statement Worksheet (Customer Support Data Analysis)

Within one week, analyze customer support ticket data to evaluate resolution efficiency and customer satisfaction across ticket categories and priority levels. Identify key performance trends by examining average resolution time, customer satisfaction scores, the percentage of tickets resolved within 24 hours, and the relationship between resolution time and customer satisfaction. Use data cleaning, exploratory data analysis, statistical techniques, and data visualization to identify performance gaps, determine key factors associated with support outcomes, and develop at least three actionable recommendations to improve customer support efficiency and customer experience.

1 Context

Customer support teams need to resolve tickets efficiently while maintaining high customer satisfaction. This project will analyze customer support ticket data to identify trends in resolution time, ticket priority, and customer satisfaction. Within one week, the analysis will generate actionable insights and recommendations to improve service efficiency and customer experience.

2 Criteria for success

Complete the project within one week by cleaning and validating the dataset, calculating key performance indicators, analyzing average resolution time by priority and customer satisfaction by category, determining the percentage of eligible closed tickets resolved within 24 hours, evaluating the relationship between resolution time and customer satisfaction, building an interactive Tableau dashboard, and presenting at least three actionable, evidence-based recommendations..

3 Scope of solution

Clean and preprocess data using Python and Pandas, perform EDA and KPI analysis, evaluate ticket performance, analyze resolution times by priority, category, and date, assess customer satisfaction and its correlation with resolution time, create Python visualizations and interactive Tableau dashboards, and deliver actionable insights and business recommendations.

4 Constraints within solution space

- The project must be completed within one week.
- Analysis is limited to the available dataset fields and records.
- Missing or inaccurate data may affect the results.
- Resolution time is assumed to be measured in hours, as the dataset does not explicitly specify the unit.The 24-hour resolution rate will be calculated based on this assumption.
- Customer satisfaction analysis is limited to tickets with valid feedback.
- Correlation does not imply causation.
- Findings may not generalize beyond the available dataset.
- Time constraints may limit advanced statistical analysis.

5 Stakeholders to provide key insight

- Customer Support Manager
- Support Agents
- Customer Experience Team
- Operations Manager
- Senior Management

6 Key data sources

Primary source: Customer Support Ticket Dataset.
Key fields: ticket_id, date, category, priority, status, resolution_time, customer_satisfaction, customer_email, agent_name.
Tools: Python (Pandas, Matplotlib, Seaborn), Tableau, and Excel or CSV files.